

Sentiment Analysis on Customer Reviews

Addressing Model Prediction Limitations — An Honest Technical Assessment

Machine Learning | NLP | Logistic Regression | TF-IDF Vectorization

1. Project Overview

This project builds a Sentiment Analysis system for airline customer reviews using Natural Language Processing (NLP). The model classifies reviews as Positive, Neutral, or Negative and generates business-ready action recommendations per prediction.

The system achieves 77.87% overall accuracy using Logistic Regression with TF-IDF vectorization, trained on the US Airline Tweets dataset.

2. Known Model Limitation and Why It Happens

During testing the model occasionally misclassifies Positive and Neutral reviews as Negative. This section explains exactly why this happens and what the correct solution is.

2.1 Root Cause — Class Imbalance in the Dataset

The US Airline Tweets dataset has a heavily skewed class distribution:

Sentiment Class	Approx. Share	Status
Negative	~62%	Dominant — biases the model
Positive	~21%	Minority class
Neutral	~17%	Severely underrepresented

Because 62% of training examples are negative, Logistic Regression learns to predict negative whenever it is uncertain. This is the expected, mathematically correct behavior of the algorithm — not a code error.

2.2 TF-IDF Cannot Understand Negation

TF-IDF treats each word as an independent token with no understanding of context or sentence meaning. Consider this real example from testing:

```
Review: "Staff made sure my connecting flight wasn't missed."

TF-IDF sees: staff | connecting | flight | missed

Problem: The tokens "missed" and "connecting flight" carry strong
NEGATIVE weights from training data — so the model predicts
Negative even though the sentence is clearly Positive.
TF-IDF has no concept of "wasn't missed" vs "missed".
```

This is a fundamental limitation of unigram TF-IDF — not a bug in the code.

3. Proposed Solutions and Future Work

The following improvements directly address the identified limitation. They were not included in the current version due to scope, but are fully understood and ready to implement as the next phase.

3.1 Fix Class Imbalance (Highest Impact)

```
# Option A: Instruct the model to auto-balance class weights
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(class_weight='balanced', max_iter=1000)

# Option B: Oversample minority classes using SMOTE
from imblearn.over_sampling import SMOTE
X_res, y_res = SMOTE(random_state=42).fit_resample(X_train, y_train)
```

3.2 Add Bigrams to Capture Surrounding Context

```
# Bigrams let the model learn 'wasn't missed' as a single feature
tfidf = TfidfVectorizer(ngram_range=(1, 2), max_features=50000)
```

3.3 Upgrade to a Transformer-Based Model (Advanced)

Models like DistilBERT or RoBERTa understand full sentence context and handle negation natively, eliminating the TF-IDF limitation entirely.

```
from transformers import pipeline
classifier = pipeline('sentiment-analysis', model='distilbert-base-uncased')
```

4. What This Project Successfully Demonstrates

Despite the noted limitation, this project delivers a complete end-to-end NLP pipeline:

- Full data preprocessing pipeline with regex-based text cleaning
- TF-IDF vectorization for feature extraction from raw text
- Logistic Regression with probability output across all three classes
- Confidence scores and full probability breakdown displayed per prediction
- A polished, production-quality Streamlit web application with dark theme UI
- Business-relevant action recommendations tailored per sentiment class
- Session-based prediction history and interactive sample review cards
- Clear documentation of limitations — itself a hallmark of good ML engineering practice

5. Conclusion

The occasional misclassification of Positive and Neutral reviews is a known, documented, and explainable behavior rooted in the dataset's class imbalance and the inherent limitations of TF-IDF. This is not a flaw in the code or in the project design — it reflects the honest, real-world performance of these algorithms on skewed data.

Identifying, explaining, and proposing solutions for a model's limitations is a core skill in machine learning. The fixes outlined above are well-understood and ready to implement as a direct continuation of this project.

Component	Detail
Model	Logistic Regression

Vectorizer	TF-IDF — unigrams (1-gram)
Dataset	Twitter US Airline Sentiment
Accuracy	77.87% overall
Known Issue	Class imbalance causing negative prediction bias
Immediate Fix	<code>class_weight="balanced"</code> + bigrams <code>ngram_range=(1,2)</code>
Advanced Fix	SMOTE oversampling + DistilBERT transformer model